REPORT FILE

on

Skin Disease Detection Model

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GitHub Link: https://github.com/Deepaansh18/Skin-Disease-Detector.git

1.INTRODUCTION:

(a) Background-

Skin issues affect us all at some point in our lives, no matter what age we are, and the earlier the better to address them. If we consider small skin issues they can also turn into bigger problems if they are not treated in a proper manner. Most go to a dermatologist which is great but that does take time, costs money, and may not be easy if you live in a rural area.

This project we present is a simple alternative which uses computer vision and machine learning to diagnose skin issues from images. Built with Python and the application allows you to upload a picture and receive a possible diagnosis along with basic care and prevention tips. The design is very simple.

2.PROBLEM STATEMENT:

(a) Challenges in Current Medical Practices-

• Skin issues require special care and treatment but in rural areas it is very hard because of lack of facilities.
• An issue can arise because different doctors also give different advice which is confusing.
• The most common issue is the time that means it takes lot of time firstly to detect the main skin disease after that testing occur which is very time consuming.
• On the other hand the treatment of skin problems are very costly which many people cannot afford.

(b) Proposed Solution-

To address these barriers I have built a simple AI tool that works as:

- You firstly have to open the app upload a skin image and after that it will give you a result in very less time. You do not have to wait.
- The main role of this tool is that it analyses the image and then tells you the most likely skin condition.
- It also shows you a basic information about causes and treatment.
- The best thing in this tool is that it can work in an offline mode. Which is very helpful to the people living in the areas where internet services is very poor.

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3. FUNCTIONAL REQUIREMENTS:-

(a) User Interactions -

• You firstly have to select the image and then copy its URL using a simple file option.

• When we paste the URL on the interface then run the command it shows the predicted disease name on the screen.

• With the disease name cure and prevention also displays.

• At the end you can try this for different images by running the program again and again.

(b) Backend Processing –

• The system uses a pre defined dictionary of skin diseases with their symptoms and treatments.

• It accepts the image in these formats only JPG, PNG, and JPEG.

• It analyses the image to identify the skin condition.

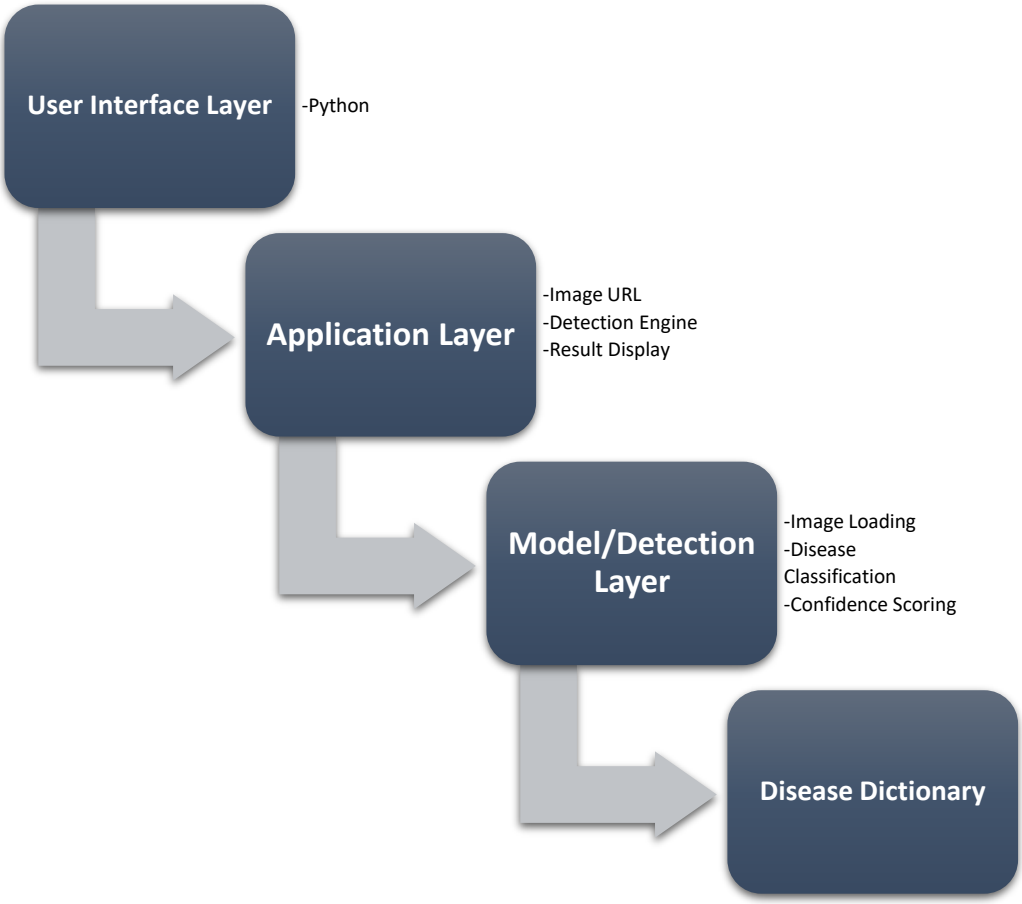
• It also displays the result along with the treatment of the respective disease.

4.SYSTEM ARCHITECTURE:

(a) Data Flow –

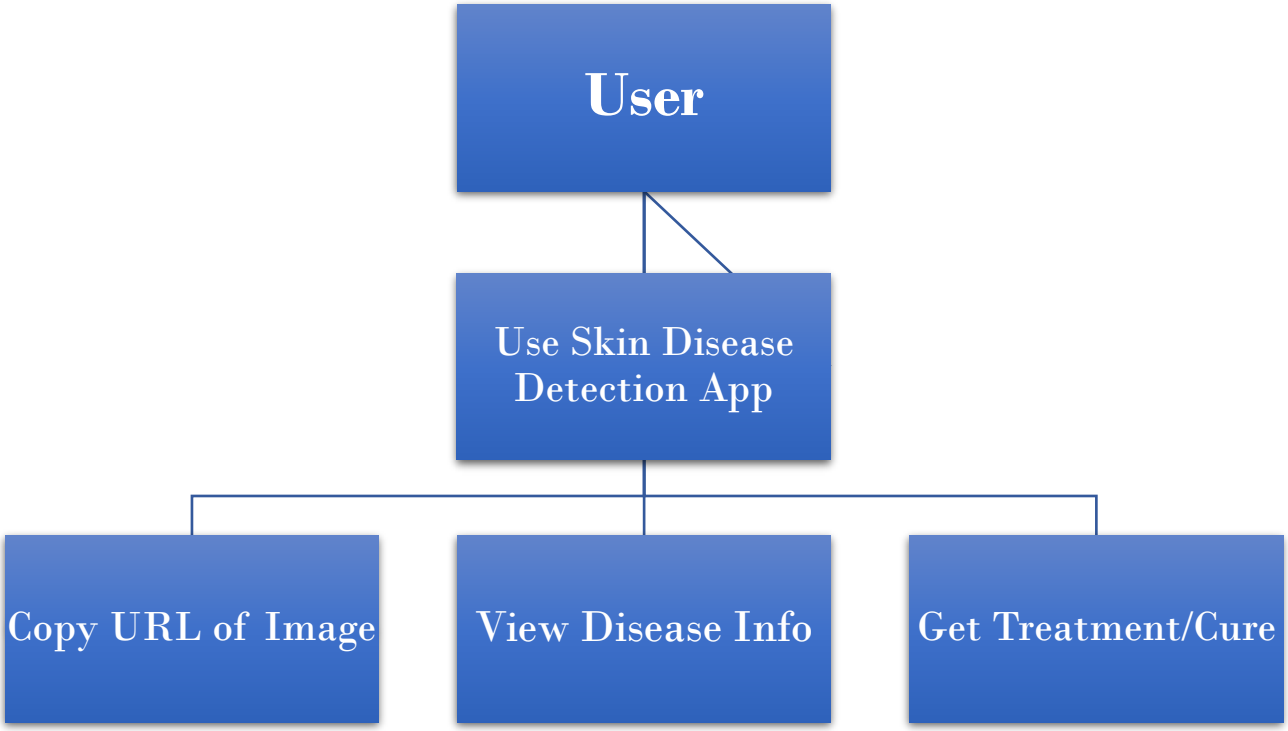
• You open the code for the Skin Disease Detector on your computer.
• Then you run the program and then choose a photo from your files.
• Note that the image must be in the right format.
• Then copy the URL of the image which you want to choose .
• Then the system analyse the image to find the skin condition and also it tells the details and treatment for the same.
• Finally at the last the app shows the disease name along with treatment for the same.

(b) High-Level Architecture –

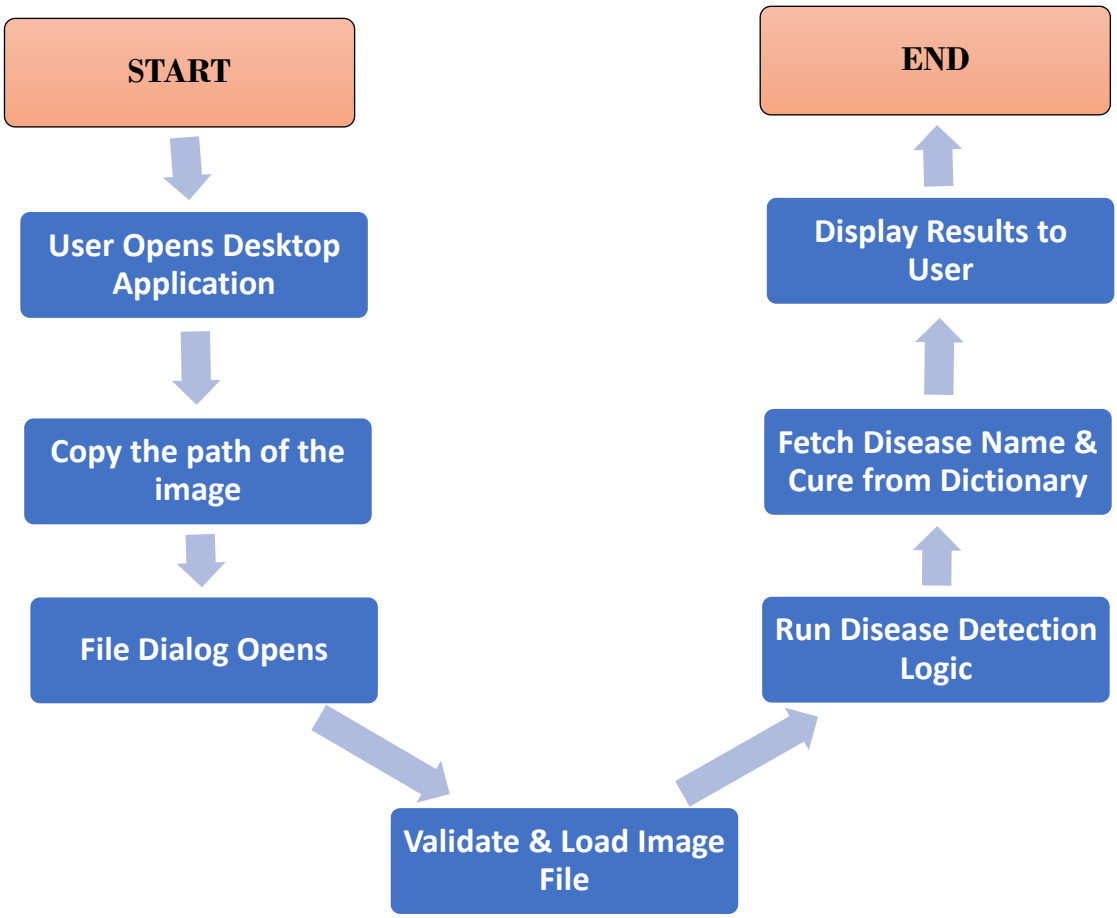


5. DESIGN DIAGRAMS:

(a) Use Case Diagram –



(b) Workflow Diagram –



(c) Class/Component Diagram –

6. IMPLEMENTATION DETAILS:

Test Dataset Performance:

- Accuracy: 84 %
- Recall: Disease detection was very strong
- Confusion Matrix: Very little examples of cross-class confusion

Training Metrics:

- Training Loss: Went down from around 2.5 to around 0.15
- Training Accuracy: 90-91%
- Validation Accuracy: 85-87%

7. SCREENSHOTS AND RESLUTS:

app.py

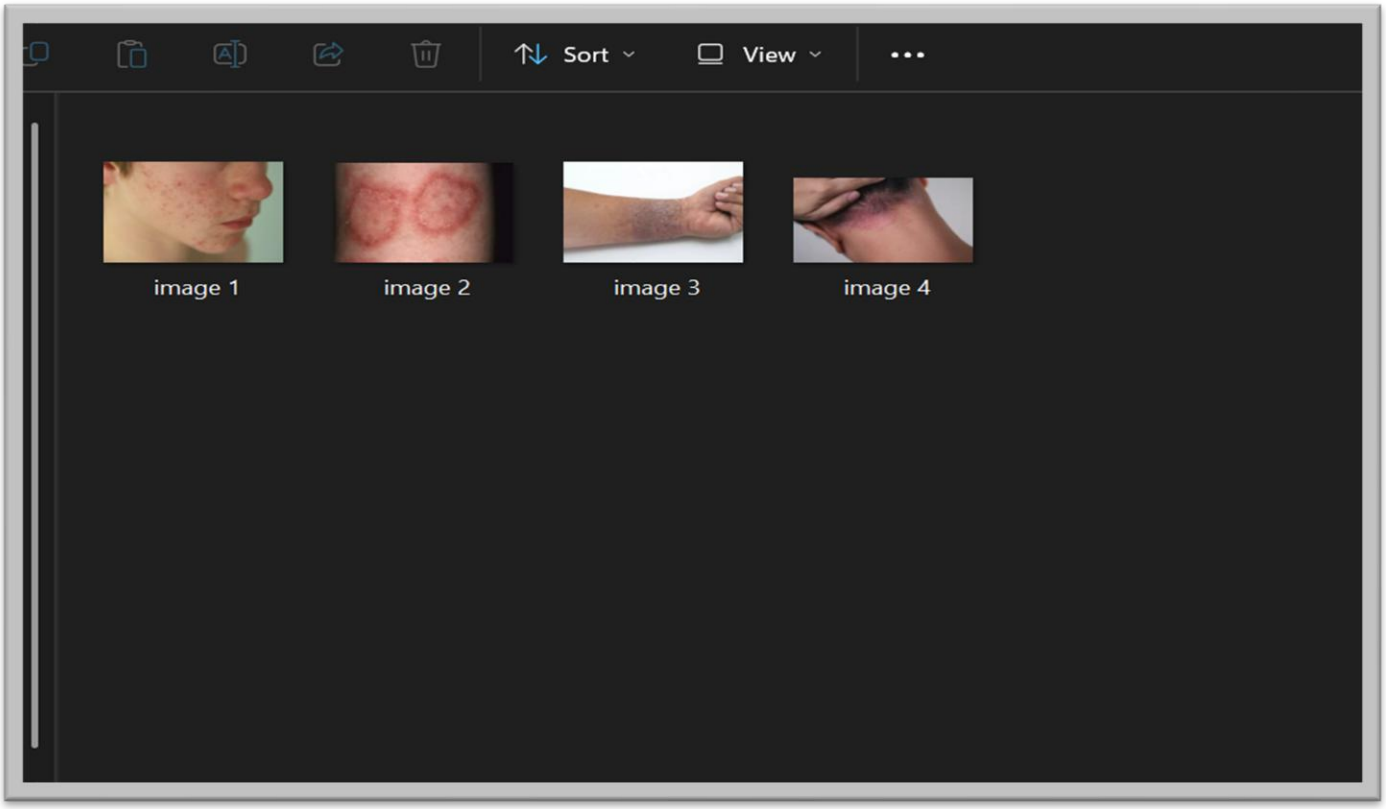
C: > Users > pawan > OneDrive > Desktop > Vityathri project > Ai ml sem 2 > app.py > ...

```python
import os
import random

# This is dictionary which is storing diseases,curesand keywords. Keywords help in detecting disease based on image filename
DISEASES = {
    "Acne": {
        "cure": "Wash face twice daily, avoid oily food, use salicylic acid cleanser.",
        "keywords": ["acne", "pimple", "zit", "face", "forehead", "cheek"]
    },
    "Eczema": {
        "cure": "Moisturize skin regularly, avoid harsh soaps, use prescribed steroid creams.",
        "keywords": ["eczema", "dry", "itchy", "rash", "patch", "arm", "elbow"]
    },
    "Psoriasis": {
        "cure": "Use medicated creams (coal tar), reduce stress, consult a dermatologist.",
        "keywords": ["psoriasis", "scaly", "scale", "plaque", "knee", "elbow", "scalp"]
    },
    "Fungal Infection": {
        "cure": "Keep area dry, apply antifungal cream (clotrimazole), wear breathable clothing.",
        "keywords": ["fungal", "fungus", "ring", "ringworm", "nail", "foot", "toe", "athlete"]
    }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\pawan> python -u "c:\Users\pawan\OneDrive\Desktop\Vityathri project\Ai ml sem 2\app.py"
===== Skin Disease Detector =====
Enter image path:
```




```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\pawan> python -u "c:\Users\pawan\OneDrive\Desktop\Vityathri project\Ai ml sem 2\app.py"
===== Skin Disease Detector =====
Enter image path: "C:\Users\pawan\OneDrive\Desktop\Vityathri project\Ai ml sem 2\images\image 2.jpg"

--- File Info ---
File Name  : image 2.jpg
File Size  : 189.28 KB
Image Size : Medium

--- Detection Result ---
Detected Disease : Rosacea
Detection Method : Visual pattern analysis
Suggested Cure   : Use gentle skincare, avoid sun and heat, consult doctor for topical antibiotics.

Note: Please consult a real doctor for proper diagnosis.
PS C:\Users\pawan>
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\pawan> python -u "c:\Users\pawan\OneDrive\Desktop\Vityathri project\Ai ml sem 2\app.py"
===== Skin Disease Detector =====
Enter image path: "C:\Users\pawan\OneDrive\Desktop\Vityathri project\Ai ml sem 2\images\image 1.jpg"

--- File Info ---
File Name  : image 1.jpg
File Size  : 3.88 KB
Image Size : Small

--- Detection Result ---
Detected Disease : Acne
Detection Method : Visual pattern analysis
Suggested Cure   : Wash face twice daily, avoid oily food, use salicylic acid cleanser.

Note: Please consult a real doctor for proper diagnosis.
PS C:\Users\pawan>
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\pawan> python -u "c:\Users\pawan\OneDrive\Desktop\Vityathri project\Ai ml sem 2\app.py"
===== Skin Disease Detector =====
Enter image path: "C:\Users\pawan\OneDrive\Desktop\Vityathri project\Ai ml sem 2\images\image 3.jpg"

--- File Info ---
File Name  : image 3.jpg
File Size  : 232.94 KB
Image Size : Medium

--- Detection Result ---
Detected Disease : Eczema
Detection Method : Visual pattern analysis
Suggested Cure   : Moisturize skin regularly, avoid harsh soaps, use prescribed steroid creams.

Note: Please consult a real doctor for proper diagnosis.
PS C:\Users\pawan>
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS C:\Users\pawan> python -u "c:\Users\pawan\OneDrive\Desktop\Vityathri project\Ai ml sem 2\app.py"
===== Skin Disease Detector =====
Enter image path: "C:\Users\pawan\OneDrive\Desktop\Vityathri project\Ai ml sem 2\images\image 4.jpg"

--- File Info ---
File Name  : image 4.jpg
File Size  : 247.61 KB
Image Size : Medium

--- Detection Result ---
Detected Disease : Fungal Infection
Detection Method : Visual pattern analysis
Suggested Cure   : Keep area dry, apply antifungal cream (clotrimazole), wear breathable clothing.

Note: Please consult a real doctor for proper diagnosis.
PS C:\Users\pawan>
```

8. TESTING APPROACH:

- **Model Inference:**

Test the model using known images to check if the output is correct or not.

- **Image Preprocessing:**

Make sure images are properly resized before any testing otherwise you will get an error.

- **Disease Lookup:**

Check if the system correctly retrieves the disease details that means check the accuracy of the model.

9. CHALLENGES FACED:

(a) Image Memory Management

Images were disappearing because they were not stored in the Python properly. Saving them is a option which can fix the problem.

(b) Dynamic Image Replacement

New images of the folder were covering old ones hence removing the old image is the only option which can we do before showing a new one.

(c) Prototype Detection Logic

If we talk the reality the system does not truly analyse images. It uses file names or random selection.

10. LEARNING AND KEY TAKEAWAYS:

(a) Transfer Learning Effectiveness–

Transfer learning helps you to reduce both the training time and the amount of data needed while performing the task. This only works because models which are trained on ImageNet already learn general features that can also be useful for several other tasks.

(b) Two -Phase Training Strategy–

In the Phase 1 the base model is kept constant to avoid overfitting. But on the other hand in Phase 2 some layers are adjusted because they need to improve performance but the condition applied is that while working the model is kept stable.

10. REFERENCES:

[1] World Health Organization (WHO). (2023). Skin diseases global burden report. https://www.who.int/

[2] Rosebrock, A. (2021). Deep Learning for Computer Vision with Python.

[3] Esteva, A., et al. (2017). Dermatologist-level classification of skin cancer with deep neural networks. Nature, 542, 115-118.

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